

Pipe and Cistern

Pipes and Cistern problems generally consist of a cistern (tank) to which one or more pipes fill the cistern or empty the cistern. These problems of pipes and cisterns can be solved by using the same method used in time and work. And we changes our formulae according to the requirement of the pipes and cisterns.

- (i) A pipe connected with a tank or a cistern that fill the tank is known as inlet.
- (ii) A pipe connected with a tank that empty it is known as outlet.

Important Points:

1. If a pipe can fill a tank in x hours, then the part filled in 1 hour = $\frac{1}{x}$
2. If a pipe can empty a tank in y hours, then the part of the full tank emptied in 1 hour = $\frac{1}{y}$
3. If a pipe can fill a tank in x hours and another pipe can empty the full tank in y hours ($y > x$), then the net part filled in 1 hour, when both the pipes are opened = $\left(\frac{1}{x} - \frac{1}{y}\right) = \frac{y-x}{xy}$

$$\text{Time taken to fill the tank} = \frac{xy}{y-x}$$

4. If a pipe can fill a tank in x hours and another pipe can fill the same tank in y hours, the part of the tank filled in 1 hour when both pipes are opened simultaneously = $\left(\frac{1}{x} + \frac{1}{y}\right) = \frac{x+y}{xy}$

$$\therefore \text{Time taken to fill completely the tank when both pipes are open simultaneously} = \frac{xy}{x+y}$$

5. If three pipes can fill a tank separately in x , y and z h respectively, then time taken to fill the tank by working together = $\frac{xyz}{xy + yz + zx}$ h.
6. If a pipe fills a tank in x hours and another fills the same tank in y hours, but a third pipe empties the full tank in z hours and all of them are opened together, the net part filled in 1 hour = $\left(\frac{1}{x} + \frac{1}{y} - \frac{1}{z}\right)$.

$$\text{Time taken to fill the tank} = \frac{xyz}{yz + xz - xy} \text{ hours}$$

Example: Pipe A can fill a water tank in 25 minutes, Pipe B can fill the tank in 40 minutes and Pipe C can empty the tank in 30 minutes. If all the 3 pipes are opened together, then in how many minutes will the tank be completely filled?

Sol. Time taken to fill the tank = $\frac{25 \times 40 \times 30}{40 \times 30 + 25 \times 30 - 25 \times 40} = \frac{600}{19} = 31\frac{11}{19}$ minutes

7. Two pipes A and B can fill a tank in x minutes and y minutes respectively. If both the pipes are opened simultaneously, Then the time after which pipe B should be closed, so that the tank is full in ' t ' minutes is

$$\left[y \left(1 - \frac{t}{x} \right) \right] \text{ minutes.}$$

Example: Two pipes A and B can fill a tank in 12 minutes and 16 minutes respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 9 minutes?

Sol. Pipe B should be closed after $= \left(1 - \frac{9}{12}\right) \times 16 = \frac{3}{12} \times 16 = 4$ min.

Types of Questions

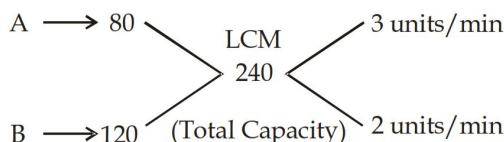
1. Pipe A can fill the tank in 80 minutes and pipe B in 120 minutes. Then after how much time both the pipes can together fill the tank?

Sol. Part filled by two taps in 1 minutes

$$= \frac{1}{80} + \frac{1}{120} = \frac{3+2}{240} = \frac{5}{240} = \frac{1}{48}$$

$\therefore$ Time taken to fill the tank = 48 minutes

Shortcut:



$\therefore$ Take LCM as the total capacity of the tank.

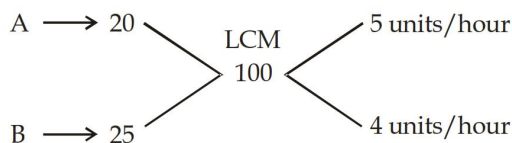
$$\text{Time taken to fill the tank} = \frac{240}{3+2} = 48 \text{ minutes}$$

2. Two pipes A and B can empty a full tank in 20 hours and 25 hours respectively. In how many hours will tank be emptied when they are opened together?

Sol. Efficiency of both pipes $= \frac{1}{20} + \frac{1}{25} = \frac{5+4}{100} = \frac{9}{100}$

$\therefore$ Tank will be emptied in $\frac{100}{9} = 11\frac{1}{9}$ hours

Shortcut:



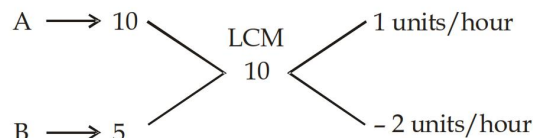
A can empty 5 units/hour and B can empty 4 units/hours.

$$\text{Time taken to empty the tank} = \frac{100}{9} = 11\frac{1}{9} \text{ hours}$$

3. A can fill a cistern in 10 hours and B can empty it in 5 hours. If they are opened together, in how many hours will cistern be empty?

Sol. Required time $= \frac{10 \times 5}{10 - 5} = \frac{10 \times 5}{5} = 10$ hours.

Shortcut:



Take negative sign when a pipe empty the tank

therefore tank will be empty in $= \frac{10}{1-2} = 10$ hours

4. Two pipes P and Q can fill a cistern in 12 and 15 min, respectively. If both are opened together and at the end of 3 min the first is closed. How much extra time will Q take to fill the cistern?

Sol. Given, time taken by P to fill the tank = 12 min.
And time taken by Q to fill the tank = 15 min.

$$\begin{aligned} \text{Part filled by both pipes in 1 min} &= \frac{1}{12} + \frac{1}{15} \\ &= \frac{5+4}{60} = \frac{9}{60} \end{aligned}$$

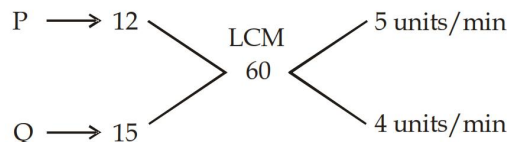
$$\text{Now, part filled in 3 min} = \frac{3 \times 9}{60} = \frac{27}{60} = \frac{9}{20}$$

$$\therefore \text{Remaining part} = 1 - \frac{9}{20} = \frac{11}{20}$$

Now, the remaining part is filled by pipe Q in x min.

$$x = \frac{11}{20} \times 15 = \frac{3 \times 11}{4} = \frac{33}{4} = 8\frac{1}{4} \text{ min}$$

Shortcut:



Total capacity = 60

Part filled by (P+Q) in 3 min $= 3(5+4) = 27$ units

Remaining part is filled by Q only

$$= \frac{60-27}{4} = \frac{33}{4} = 8\frac{1}{4} \text{ min}$$

5. A tank can be filled by two pipes A and B in 20 min and 30 min, respectively. When the tank was empty the two pipes were opened. After some time, the first pipe was stopped and the tank was filled in 18 min. After how much time of the start was the first pipe stopped?

Sol. Given, tank filled by A=20 min
And tank filled by B=30 min
Let the first pipe be closed after x min.

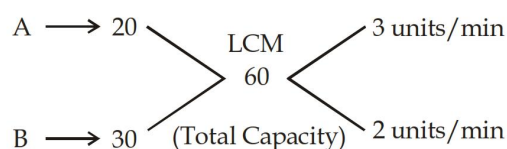
Then, work done by first pipe in x min = $\frac{x}{20}$

According to the question,

$$\frac{x}{20} + \frac{18}{30} = 1 \Rightarrow \frac{x}{20} = 1 - \frac{18}{30} \Rightarrow \frac{x}{20} = 1 - \frac{3}{5} = \frac{2}{5}$$

$$\therefore x = \frac{2}{5} \times 20 = 8 \text{ min}$$

Shortcut:



B work all time, therefore part filled by B in 18 min
= $2 \times 18 = 36$ units

Remaining part is filled by A in $\frac{24}{3} = 8$ min

Therefore, after 8 min the first pipe is closed.

6. Pipe A can fill a cistern in 20 minutes and pipe B can fill it in 30 minutes. While pipe C can empties the full cistern in 1 hours. If all the pipes be turned on at the same time, in how much time will the cistern be full?

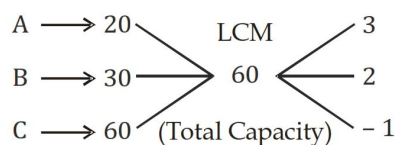
Sol. Filled part of cistern in 1 minute, when A, B and C be

$$\text{turned on at same time} = \frac{1}{20} + \frac{1}{30} - \frac{1}{60} = \frac{3+2-1}{60}$$

$$= \frac{4}{60} = \frac{1}{15}$$

$\therefore$ time taken to fill the cistern = 15 minutes

Shortcut:



Therefore Cistern will be filled in

$$\frac{60}{3+2-1} = \frac{60}{4} = 15 \text{ minutes}$$

7. Pipe A can fill a cistern in 5 hours and B can fill it in 20 hours. Both pipes are turned on but there is a leakage in the bottom of the cistern. So the cistern is filled in 30 minutes more. In how many time will leakage empty the full cistern?

Sol. Efficiency of pipe A and B = $\frac{1}{5} + \frac{1}{20} = \frac{4+1}{20} = \frac{5}{20} = \frac{1}{4}$

$\therefore$ Pipe A and B working together will fill the cistern in 4 hours

But time taken with leakage = 4 hours + 30 minutes

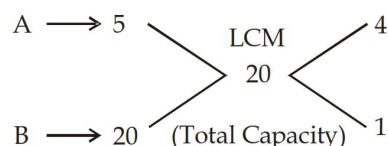
$$= \left(4 + \frac{1}{2}\right) \text{ hours} = \frac{9}{2} \text{ hours}$$

$$\frac{1}{5} + \frac{1}{20} - \frac{1}{x} = \frac{2}{9} \Rightarrow \frac{1}{5} + \frac{1}{20} - \frac{2}{9}$$

$$\frac{1}{x} = \frac{36+9-40}{180} = \frac{5}{180} = \frac{1}{36} \Rightarrow x = 36 \text{ hours}$$

$\therefore$ Leakage will empty the cistern in 36 hours

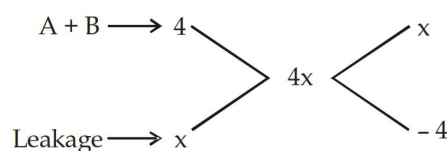
Shortcut:



A and B fill the cistern in $\frac{20}{5} = 4$ hours

But when there is a leakage in the bottom then the

tank is filled in $\left(4 + \frac{1}{2}\right) = \frac{9}{2}$ hours



$$\frac{4x}{x-4} = \frac{9}{2} \Rightarrow x = 36 \text{ hours}$$

8. Two pipes can fill a cistern in 20 minutes and 25 minutes respectively. These are opened alternatively for 1 minutes each, beginning with first. In how much time will the cistern be full?

Sol. Filled cistern in 1 round (or 2 minutes)

$$= \frac{1}{20} + \frac{1}{25} = \frac{5+4}{100} = \frac{9}{100}$$

$$\left(\frac{9}{100} \Rightarrow 100 \div 9 \text{ gives 11 round complete}\right)$$

$\therefore$ Filled cistern in 11 round (or 22 minutes)

$$= \frac{9}{100} \times 11 = \frac{99}{100}$$

$$\therefore \text{Empty part of cistern} = 1 - \frac{99}{100} = \frac{1}{100}$$

On 23rd min., it is first pipe's is opened.

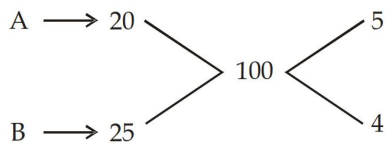
$\therefore$ Time taken to fill empty part by first pipe

$$= \frac{\text{empty part}}{\text{efficiency of first pipe}} = \frac{\frac{1}{100}}{\frac{1}{20}}$$

$$= \frac{1}{100} \times \frac{20}{1} = \frac{1}{5} \text{ minutes}$$

$$\therefore \text{Total time} = 22 + \frac{1}{5} = 22\frac{1}{5} \text{ minutes}$$

Shortcut:



A and B opened alternatively.

$\therefore$ In 2 minutes filled is 9 units.

$$\begin{array}{cc} \text{Minutes} & \text{Tank Filled} \\ \times 11 \left(\begin{array}{cc} 2 & 9 \\ 22 & 99 \end{array} \right) \times 11 \end{array}$$

Remaining part = $100 - 99 = 1$ unit

1 unit is filled by A in $= \frac{1}{5}$ minutes

$$\text{Total time} = 22 + \frac{1}{5} = 22\frac{1}{5} \text{ minutes}$$

9. A swimming pool has 3 drain pipes. The first two pipes A and B, operating simultaneously can empty the pool in half the time that C, the 3rd pipe, alone takes to empty it. Pipe A, working alone, takes half the time taken by pipe B. Together they take 6 h 40 min to empty the pool. Time taken by pipe A to empty the pool, in hours, is

Sol. Let the time taken by pipe B = $2x$ h

$$\text{Part of the pool emptied by B in 1 h} = \frac{1}{2x}$$

$$\text{Part of pool emptied in 1 h by A} = \frac{1}{x}$$

$\therefore$ Time taken by pipe

$$C = \frac{2}{\frac{1}{2x} + \frac{1}{x}} = \frac{2}{\frac{1+2}{2x}} = \frac{4x}{3} \text{ h}$$

Now, part of the tank filled by all together

$$= \frac{1}{x} + \frac{1}{2x} + \frac{3}{4x} = \frac{4+2+3}{4x} = \frac{1}{6+\frac{2}{3}}$$

$$= \frac{4+2+3}{4x} = \frac{3}{20} = 9 \times 20 = 4x \times 3 \Rightarrow x = \frac{9 \times 20}{4 \times 3} = 15 \text{ h}$$

10. A tank has a leak which would empty the completely filled tank in 10 h. If the tank is full of water and a tap is opened which admits 4 L of water per minutes in the tank, the leak takes 15 h to empty the tank. How many litres of water does the tank hold?

Sol. Let the capacity of the tank = x L

According to the question,

$$\text{Quantity of water emptied by the leak in 1 h} = \frac{x}{10} \text{ L}$$

$$\text{Quantity of water emptied with inlet pipe in 1 h} = \frac{1}{15}$$

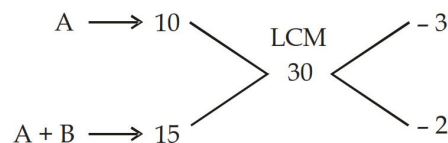
Let Inlet pipe fills tank in x h

$$\text{So, } \frac{1}{x} - \frac{1}{10} = -\frac{1}{15} \Rightarrow \frac{1}{x} = \frac{1}{10} - \frac{1}{15} \Rightarrow \frac{3-2}{30} = \frac{1}{30}$$

So, Inlet pipe fills tank in 30 h

Capacity of tank = $30 \times 60 \times 4 = 7200$ L.

Shortcut:



Therefore B admit 1 unit/hour and originally B admit 4 L/min. Therefore

$$1 = 4 \times 60 \Rightarrow 1 \equiv 240$$

$$\text{Total Capacity} = 30 \times 240 = 7200 \text{ L}$$

Foundation

Questions

1. If a pipe fills a tank in 6 h, then what part of the tank will the pipe fill in 1 h?

(a) $\frac{1}{3}$
(b) $\frac{1}{6}$

(c) $\frac{1}{4}$
(d) $\frac{1}{5}$
2. An inlet pipe fills $\frac{1}{8}$ part of a tank in 1 h. How much time will the pipe take to fill the empty tank?

(a) 4h
(b) 2h

(c) 6h
(d) 8h
3. An outlet pipe can empty a cistern in 3 h. In what time will the pipe empty two-third part of the cistern?

(a) 4h
(b) 2h

(c) 3h
(d) 5h
4. There are two taps A and B to fill up a water tank. The tank can be filled in 40 min, if both taps are on. The same tank can be filled in 60 min, if tap A alone is on. How much time will tap B alone take, to fill up the same tank?

(a) 64 min
(b) 80 min

(c) 96 min
(d) 120 min
5. A pipe can fill a tank in 10 h, while an another pipe can empty it in 6 h. Find the time taken to empty the tank, when both the pipes are opened simultaneously?

(a) 11h
(b) 15h

(c) 18h
(d) 16h
6. Three taps are fitted in a cistern. The empty cistern is filled by the first and the second taps in 3 and 4h, respectively. The full cistern is emptied by the third tap in 5 h. If all three taps are opened simultaneously, the empty cistern will be filled up in?

(a) $1\frac{14}{23}$ h
(b) $2\frac{14}{23}$ h

(c) 2 h 40 min
(d) 1 h 56 min
7. Pipe A can fill a tank in 30 min, while pipe B can fill the same tank in 10 min and pipe C can empty the full tank in 40 min. If all the pipes are opened together, how much time will be needed to make the tank full?

(a) $9\frac{3}{13}$ h
(b) $9\frac{4}{13}$ h

(c) $9\frac{7}{13}$ h
(d) $9\frac{9}{13}$ h
8. Three taps A, B and C together can fill an empty cistern in 10 min. The tap A alone can fill it in 30 min and the tap B alone can fill it in 40 min. How long will the tap C alone take to fill it?

(a) 16 min
(b) 24 min

(c) 32 min
(d) 40 min
9. Two pipes A and B can fill a tank in 1 h and 75 min, respectively. There is also an outlet C. If all the three pipes are opened together. The tank is full in 50 min. How much time will be taken by C to empty the full tank?

(a) 100 min
(b) 150 min

(c) 200 min
(d) 125 min
10. A, B and C are three pipes connected to a tank. A and B together fill the tank in 6 h, B and C together fill the tank in 10 h and A and C together fill the tank in 12 h. In how much time A, B and C fill up the tank together?

(a) 9 h
(b) $5\frac{3}{7}$ h

(c) $5\frac{2}{7}$ h
(d) $5\frac{5}{7}$ h
11. Inlet A is four times faster than inlet B to fill a tank. If A alone can fill it in 15 min, how long will it take if both the pipes are opened together?

(a) 10 min
(b) 12 min

(c) 15 min
(d) 14 min
12. There are two inlets A and B connected to a tank. A and B can fill the tank in 16 h and 10 h, respectively. If both the pipes are opened alternately for 1 h, starting from A, then how much time will the tank take to be filled?

(a) $13\frac{1}{4}$ h
(b) $11\frac{6}{8}$ h

(c) $12\frac{2}{5}$ h
(d) $12\frac{1}{4}$ h
13. A pipe can empty a cistern in 27 hours. Find the time in which $\frac{2}{3}$ part of the cistern will be emptied?

(a) 9 hours
(b) 12 hours

(c) 15 hours
(d) 18 hours
14. A water tank is $\frac{2}{3}$ rd full. Pipe A can fill the tank in 10 minutes and the pipe B can empty it in 6 minutes. If both the pipes are open, how long will it take to empty or fill the tank completely?

(a) 6 minutes to empty
(b) 6 minutes to fill

(c) 9 minutes to empty
(d) 10 minutes to empty

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15. A tap can fill a cistern in 8 hours and another can empty it in 16 hours. If both the taps are opened simultaneously, the time (in hours) to fill the tank is?
 (a) 8 (b) 10
 (c) 16 (d) 24
16. A pipe can empty a tank in 15 hrs and another pipe can empty it in 10 hours. If both the pipes are opened simultaneously, find the time in which a full tank is emptied?
 (a) 8 hrs (b) 6 hrs
 (c) 4 hrs (d) 5 hrs
17. Two pipes A and B can fill a cistern in 20 minutes and 25 minutes respectively. Both are opened together, but after 5 minutes, B is turned off. How much longer will the cistern take to fill?
 (a) 16 minutes (b) 18 minutes
 (c) 11 minutes (d) C.N.D.
18. 12 pumps working 6 hours a day can empty a completely filled reservoir in 15 days. How many such pumps working 9 hours a day will empty the same reservoir in 12 days?
 (a) 15 (b) 9
 (c) 10 (d) 12
19. A tank has a leak which would empty it in 8 hrs. A tap is turned on which admits 6 litres a minutes into the tank, and it is now emptied in 12 hrs. How many litres does the tank hold?
 (a) 8260 litres (b) 8450 litres
 (c) 8640 litres (d) 8660 litres
20. A tap can fill the cistern in 8 hours and another can empty it in 16 hours. If both the taps are opened simultaneously, the time (in hours) to fill the tank is?
 (a) 16 hrs (b) 8 hrs
 (c) 10 hrs (d) 12 hrs
21. A pipe can fill a tank with water in 5 hours and an another pipe can empty the same tank in 4 hours. If the tank is completely filled and both the taps are opened together than the tank will be empty in following time (hour)?
 (a) 9 (b) 18
 (c) 20 (d) $20\frac{1}{2}$
22. Two pipes can fill a tank respectively in 15 and 12 hours while a third pipe can empty it in 20 hours. If the tank is empty and all three pipes are opened together then in how much time will it take to fill the tank completely?
 (a) 10 hours (b) 5 hours
 (c) 15 hours (d) 20 hours
23. 2 pipes A and B can fill a tank separately in 60 min and 70 min respectively. There is a third pipe attached to the bottom of the tank to empty it. The tank is filled in 60 min when all three pipes are opened. In how much time (min) the third pipe alone take to fill the tank?
 (a) 110 min (b) 70 min
 (c) 120 min (d) 90 min
24. Two pipes A and B can fill a tank in $37\frac{1}{2}$ min and 45 min respectively. If both the pipes are opened together, then after how much time should B be closed so that the tank is full in half hours?
 (a) 15 min (b) 10 min
 (c) 21 min (d) 9 min
25. A big tanker can be filled by two pipes A and B in 60 and 40 min respectively. Then what time it will take to fill an empty tanker if tap B is used half of time and tap A and B together are used rest half of time?
 (a) 24 min (b) 30 min
 (c) 18 min (d) 32 min
26. Two taps A and B can fill an oil tank in 15 min and 18 min respectively while a third tap is used to empty the tank. The third tap is opened after 6 min of opening of tap A and B. If the tank empties in $16\frac{1}{2}$ min after opening the third tap then in how much time the third tap can empty the full tank?
 (a) 6 min (b) 12 min
 (c) 10 min (d) 15 min
27. Three pipes A, B and C can fill a tank in 6 hours. After running them together for two hours the tap C is closed and A and B fills it completely in 7 hours. How much time would C take to fill the tank alone?
 (a) 7 hours (b) 10.5 hours
 (c) 14 hours (d) 21 hours
28. A water tank has three taps A, B, C. Tap A, when opened, can fill the water tank alone in 4 hours. Tap B, when opened, can filled the water tank alone in 6 hours and tap C, when opened, can empty the water tank alone in 3 hours. If taps A, B and C are opened simultaneously how long will it take to fill the tank completely?
 (a) 10 hours (b) 8 hours
 (c) 18 hours (d) 12 hours
29. A water tank has two taps A and B. A can fill it in 6 hours, and tap B empty it in 5 hours. If Both the taps open simultaneously the time required to empty the tank?
 (a) 20 hours (b) 30 hours
 (c) 25 hours (d) 35 hours
30. 18 pumps can fill reservoir of capacity 1440 kilo liter in 5 hours. Then how many hours required to fill a reservoir of capacity 1920 kilo liter by 8 pumps?
 (a) 10 hours (b) 16 hours
 (c) 15 hours (d) 20 hours
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Moderate

1. Two pipes can fill a cistern in 14 h and 16 h, respectively. The pipes are opened simultaneously and it is found that due to leakage in the bottom, it took 92 min more to fill the cistern. When the cistern is full, in what time will the leak empty it?

(a) $43\frac{15}{23}$ h (b) $43\frac{17}{23}$ h

(c) $43\frac{13}{23}$ h (d) $43\frac{19}{23}$ h
2. A pipe P can fill a tank in 12 min and another pipe R can fill it in 15 min. But, the 3rd pipe M can empty it in 6 min. The 1st two pipes P and R are kept open for 5 min in the beginning and then the 3rd pipe is also opened. In what time is the tank emptied?

(a) 30 min (b) 25 min

(c) 45 min (d) 35 min
3. A tap having diameter 'd' can empty a tank in 40 min. How long another tap having diameter '2d' take to empty the same tank?

(a) 5 min (b) 20 min

(c) 10 min (d) 40 min
4. If two pipes function together, the tank will be filled in 12 h. One pipe fills the tank in 10 h faster than the other. How many hours does the faster pipe take to fill up the tank?

(a) 20 h (b) 60 h

(c) 25 h (d) 35 h
5. Two pipes A and B can fill a cistern in 15 and 20 min, respectively. Both the pipes are opened together, but after 2 min, pipe A is turned off. What is the total time required to fill the tank?

(a) $\frac{46}{3}$ min (b) $\frac{52}{3}$ min

(c) $\frac{43}{3}$ min (d) $\frac{41}{3}$ min.
6. Two pipes A and B can fill a tank in 24 and 32 min, respectively. If both the pipes are opened together, after how much time pipe B should be closed so that the tank is full in 9 min?

(a) 40 min (b) 30 min

(c) 10 min (d) 20 min
7. Two taps A and B can fill a tank in 20 min and 30 min, respectively. An outlet pipe C can empty the full tank in 15 min. A, B and C are opened alternatively, each for 1 min. How long will the tank take to be filled?

(a) 105 min (b) 120 min

(c) 167 min (d) 185 min
8. Two taps A and B can fill a tank in 25 min and 20 min, respectively. But taps are not opened properly, so the taps A and B allow $\frac{5}{6}$ th and $\frac{2}{3}$ rd part of water, respectively. How long will they take to fill the tank?

(a) 12 min (b) 13 min

(c) 14 min (d) 15 min
9. Three taps A, B and C fill a tank in 20 min, 15 min and 12 min, respectively. If all the taps are opened simultaneously, how long will they take to fill 40% of the tank?

(a) 1 min (b) 2 min

(c) 3 min (d) 4 min
10. A pipe can fill a cistern in 4 min and another pipe can fill it in 5 min, but a third pipe can empty it in 2 min. The first two pipes are kept open for 2 min in the beginning and then the third pipe is also opened. Time taken to empty the cistern is?

(a) 20 min (b) 22 min

(c) 42 min (d) 18 min
11. Taps A, B and C attached with a tank and velocity of water coming through them are 42 L/h, 56 L/h and 48 L/h, respectively. A and B are inlets and C is outlet. If all the taps are opened simultaneously, tank is filled in 16 h. What is the capacity of the tank?

(a) 2346 L (b) 1600 L

(c) 800 L (d) 960 L
12. A boy and girls together fill a cistern with water. The boy pours 4 litres of water every 3 minutes and the girl pours 3 litres every 4 minutes. How much time will it take to fill 100 litres of water in the cistern?

(a) 36 minutes (b) 42 minutes

(c) 48 minutes (d) 44 minutes
13. A tap takes 36 hours extra to fill a tank due to a leakage equivalent to half of its inflow. The inlet pipe alone can fill the tank in how many hours?

(a) 36 (b) 24

(c) 30 (d) 18
14. A tank be filled by three pipes whose diameter are 1 cm, 3 cm and 4 cm running together if all three pipes fill the tank in 16 minutes. find out in what time the third pipe alone will fill the tank. If amount of water flowing from each pipe being proportional to square of its diameter?

(a) 26 minutes (b) 416 minutes

(c) 24 minutes (d) 312 minutes
15. Two pipes A and B can fill a tank in 36 hours and 45 hours respectively. If both the pipes are opened simultaneously, how much time will be taken to fill the tank?

- (a) 10 hrs (b) 15 hrs
(c) 20 hrs (d) 25 hrs
16. In what time would a cistern be filled by three pipes whose diameters are 1 cm, $1\frac{1}{3}$ cm, 2 cm, running together, when the largest alone will fill it in 61 minutes, the amount of water flowing in by each pipe being proportional to the square of its diameter?
(a) 32 minutes (b) 34 minutes
(c) 36 minutes (d) 38 minutes
17. Two pipes can separately fill a tank in 20 hrs and 30 hrs respectively. Both the pipes are opened to fill the tank but when the tank is $\frac{1}{3}$ full a leak develops in the tank through which $\frac{1}{3}$ of the water supplied by both the pipes per hour leak out. What is the total time taken to fill the tank?
(a) 12 hrs (b) 14 hrs
(c) 16 hrs (d) 18 hrs
18. A cistern can be filled by two pipes A and B in 4 hours and 6 hours respectively. When full, the tank can be emptied by a third pipe C in 8 hours. If all the taps be turned on at the same time, the cistern will be full in?
(a) $2\frac{2}{7}$ hrs (b) $3\frac{3}{7}$ hrs
(c) $4\frac{3}{7}$ hrs (d) $5\frac{5}{7}$ hrs
19. If two pipes function simultaneously, the reservoir will be filled in 6 hours. One pipe fills the reservoir 5 hours faster than the other. How many hours does the faster pipe take to fill the reservoir?
(a) 8 hrs (b) 10 hrs
(c) 12 hrs (d) C.N.D.
20. Three pipes A, B and C can fill cistern in 6 hrs. After working together for 2 hours, C is closed and A and B fill the cistern in 8 hrs. Then find the time in which the cistern can be filled by pipe C?
(a) 6 hrs (b) 12 hrs
(c) 14 hrs (d) 20 hrs

Difficult

1. Two pipes A and B can fill a tank in 24 minutes and 32 minute respectively. If both the pipes are opened simultaneously, after how much time should B be closed so that the tank is full in 18 minutes?
(a) 2 min. (b) 4 min.
(c) 6 min. (d) 8 min.
2. If three taps are opened together, a tank is filled in 12 hrs. One of the taps can fill it in 10 hrs and another in 15 hrs. How many hours does the third tap take alone to empty it completely?
(a) 8 hrs (b) 10 hrs
(c) 12 hrs (d) 16 hrs
3. A tank can be filled with water by two pipes A and B together in 36 minutes. If the pipe B was closed after 30 minutes, the tank is filled in 40 minutes. The pipe B can alone fill the tank in?
(a) 45 minutes (b) 60 minutes
(c) 75 minutes (d) 90 minutes
4. A tank has two pipes. The first pipe can fill it in 45 minutes and the second can empty it in 1 hour. In what time will the empty tank be filled if the pipes be opened one at a time in alternate minutes?
(a) 2 hrs 55 min (b) 3 hrs 40 min
(c) 4 hrs 48 min (d) 5 hrs 53 min
5. A tank can be filled by a tap in 20 min and by another tap in 60 min. Both the taps are kept open for 5 min and then the 1st tap is shut off. After this, how much time the tank will be completely filled?
(a) 20 min (b) 30 min
(c) 45 min (d) 40 min
6. A cistern has three pipes A, B and C. Pipes A and B can fill it in 3 h and 4 h, respectively, while pipe C can empty the completely filled cistern in 1 h. If pipes are opened in order at 3:00 pm, 4:00 pm and 5:00 pm, respectively, at what time will the cistern be empty?
(a) 6 : 15 pm (b) 7 : 12 pm
(c) 8 : 12 pm (d) 8 : 35 pm
7. Three pipes A, B and C can fill a tank in 30 min, 20 min and 10 min, respectively. When the tank is empty, all the three pipes are opened. If A, B and C discharge chemical solutions P, Q and R respectively, then the part of solution R in the liquid in the tank after 3 min is:
(a) $\frac{8}{11}$ (b) $\frac{5}{11}$
(c) $\frac{6}{11}$ (d) $\frac{7}{11}$
8. Three pipes A, B and C can fill a cistern in 6 hrs. After working together for 2 hours, C is closed and A and B fill the cistern in 8 hrs. Then find the time in which the cistern can be filled by pipe C?
(a) 14 hrs. (b) 12 hrs.
(c) 16 hrs. (d) 18 hrs.

9. A, B and C are three pipes connected to tank. A and B together fill the tank in 6 hrs. B and C together fill the tank in 10 hrs. A and C together fill the tank in $7\frac{1}{2}$ hrs. In how much time will A, B and C fill the tank together?
 (a) 36 hrs. (b) 32 hrs.
 (c) 30 hrs. (d) 5 hrs.
10. Two pipes A and B can fill a tank in 15 hours and 20 hours respectively while a third pipe C can empty the full tank in 25 hours. All the three pipes are opened in the beginning. After 10 hours C is closed. Find, in how much time will the tank be full?
 (a) 12 hrs (b) 8 hrs
 (c) 10 hrs (d) 14 hrs
11. Three pipes A, B and C can fill a cistern in 10 hours, 12 hours and 15 hours respectively. First A was opened. After 1 hour, B was opened and after 2 hours from the start of A, C was also opened. Find the time in which the cistern is just full?
 (a) 2 hrs (b) 4 hrs
 (c) 2 hrs 52 min (d) 4 hrs 52 min
12. A, B, C are pipes attached to a cistern. A and B can fill it in 20 and 30 minutes respectively, while C can empty it in 15 minutes. If A, B, C be kept open successively for 1 minute each, how soon will the cistern be filled?
 (a) 167 min (b) 160 min
 (c) 166 min (d) 164 min
13. A tank is filled in 5 hours by three pipes A, B and C. The pipe C is twice as fast as B and B is twice as fast as A. How much time will pipe A alone take to fill the tank?
 (a) 20 hrs (b) 25 hrs
 (c) 35 hrs (d) 15 hrs
14. Two pipes A and B can fill a tank in 15 minutes and 20 minutes respectively. Both the pipes are opened together but after 4 minutes, pipe A is turned off. What is the total time required to fill the tank?
 (a) 10 min 20 sec (b) 11 min 45 sec
 (c) 12 min 30 sec (d) 14 min 40 sec
15. Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternately, the tank will be full in?
 (a) 6 hrs (b) $6\frac{2}{3}$ hrs
 (c) 5 hrs (d) 7 hrs
16. A booster pump can be used for filling as well as for emptying a tank. The capacity of the tank is 2400 m^3 . The emptying capacity of the tank is 10 m^3 per minute higher than its filling capacity and the pump needs 8 minutes lesser to empty the tank than it needs to fill it. What is the filling capacity of the pump?
 (a) $50\text{ m}^3/\text{min}$ (b) $60\text{ m}^3/\text{min}$
 (c) $72\text{ m}^3/\text{min}$ (d) $32\text{ m}^3/\text{min}$
17. A leak in the bottom of a tank can empty the full tank in 8 hours. An inlet pipe fills water at the rate of 6 litres a minute. When the tank is full, the inlet pipe is opened and due to the leak, the tank is empty in 12 hours. How many litres does the tank hold?
 (a) 7580 (b) 7960
 (c) 8290 (d) 8640
18. Two pipes can fill a tank in 20 and 24 minutes respectively and a waste pipe can empty 3 gallons per minute. All the three pipes working together can fill the tank in 15 minutes. The capacity of the tank is?
 (a) 60 gallons (b) 100 gallons
 (c) 120 gallons (d) 180 gallons
19. Three pipes A, B and C can fill a tank in 6 hours. After working at it together for 2 hours, C is closed and A and B can fill the remaining part in 7 hours. The number of hours taken by C alone to fill the tank is?
 (a) 10 (b) 12
 (c) 14 (d) 20
20. Two pipes A and B can fill a tank in 24 min. and 48 min. respectively. If both the pipes are opened simultaneously, after how much time B should be closed so that the tank is full in 20 minutes?
 (a) 8 (b) 12
 (c) 14 (d) 16

Previous Year (Memory Based)

1. Three taps A, B and C can fill a tank in 12, 15 and 20 hours respectively. If A is open all the time and B and C are open for one hour each alternatively, the tank will be full in:
 (a) 6 hours (b) $6\frac{1}{2}$ hours
 (c) 7 hours (d) $7\frac{1}{2}$ hours
2. A pipe of diameter 'd' can drain a certain water tank in 40 minutes. The time taken by a pipe of diameter '2d' for doing the same job is:
 (a) 5 minutes (b) 10 minutes
 (c) 20 minutes (d) 80 minutes
3. A pipe can fill a tank with water in 3 hours. Due to a leakage in bottom, it takes $3\frac{1}{2}$ hours to fill it. In what time the leak will empty the fully filled tank?

-
- | | |
|--|---|
| <p>(a) 12 hours (b) 21 hours</p> <p>(c) $6\frac{1}{2}$ hours (d) $10\frac{1}{2}$ hours</p> | <p>(a) $(x - y)$ hours (b) $(y - x)$ hours</p> <p>(c) $\frac{xy}{x - y}$ hours (d) $\frac{xy}{y - x}$ hours</p> |
|--|---|
4. Two pipes A and B can separately fill a cistern in 60 minutes and 75 minutes respectively. There is a third pipe in the bottom of the cistern to empty it. If all the three pipes are simultaneously opened, then the cistern is full in 50 minutes. In how much time the third pipe alone can empty the cistern?

(a) 110 minutes (b) 100 minutes

(c) 120 minutes (d) 90 minutes
 5. A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. What is the total time taken to fill the tank completely?

(a) 4 hrs (b) 4 hrs 15 min.

(c) 3 hrs 15 min. (d) 3 hrs 45 min.
 6. One pipe can fill a tank three times as fast as another pipe. If together the two pipes can fill the tank in 36 minutes, the slower pipe alone will be able to fill tank in

(a) 81 minutes (b) 108 minutes

(c) 144 minutes (d) 192 minutes
 7. A cistern has two pipes. One can fill it with water in 8 hours and other can empty it in 5 hours. In how many hours will the cistern be emptied if both the pipes are opened together when $\frac{3}{4}$ of the cistern is already full of water?

(a) $13\frac{1}{3}$ hours (b) 10 hours

(c) 6 hours (d) $3\frac{1}{3}$ hours
 8. A tank is fitted with two taps. The first tap can fill the tank completely in 45 minutes and the second tap can empty the full tank in one hour. If both the taps are opened alternately for one minute, then in how many hours the empty tank will be filled completely?

(a) 2 hours 55 minutes (b) 3 hours 40 minutes

(c) 4 hours 48 minutes (d) 5 hours 53 minutes
 9. Two pipes can fill a tank with water in 15 and 12 hours respectively and a third pipe can empty it in 4 hours. If the pipes be opened in order at 8, 9 and 11 a.m. respectively, the tank will be emptied at

(a) 11 : 40 a.m. (b) 12 : 40 p.m.

(c) 1 : 40 p.m. (d) 2 : 40 p.m.
 10. A pipe can fill a tank ' x ' hours and another pipe can empty it in ' y ' ($y > x$) hours. If both the pipes are open. In how many hours will the tank be filled?
 11. 12 pumps working 6 hours a day can empty a completely filled reservoir in 15 days. How many such pumps working 9 hours a day will empty the same reservoir in 12 days?

(a) 15 (b) 9

(c) 10 (d) 12
 12. A tap takes 36 hours extra to fill a tank due to a leakage equivalent to half of its inflow. The inflow can fill the tank in how many hours?

(a) 36 hrs (b) 24 hrs

(c) 30 hrs (d) 18 hrs
 13. A tank can be filled with water by two pipes, A and B together in 36 minutes. If the pipe B was stopped after 30 minutes, the tank is filled in 40 minutes. The pipe B can alone fill the tank in:

(a) 45 minutes (b) 60 minutes

(c) 75 minutes (d) 90 minutes
 14. Two pipes A and B fill a water tank in 20 and 24 minutes respectively and a third pipe C can empty at the rate of 3 gallons per minute. If A, B and C are opened together to fill the tank in 15 minutes, find the capacity of tank?

(a) 180 (b) 150

(c) 120 (d) 60
 15. Three pipes P, Q and R can separately fill a cistern in 4, 8 and 12 hours respectively. Another pipe S can empty the completely filled cistern in 10 hours. Which of the following arrangements will fill the empty cistern in less time than others?

(a) Q alone is open (b) P, R and S are open

(c) P and S are open (d) P, Q and S are open
 16. A tank has a leak which would empty the completely filled tank in 10 hours. If the tank is full of water and a tap is opened which admits 4 litres of water per minute in the tank, the leak takes 15 hours to empty the tank. How many litres of water does the tank hold?

(a) 2400 L (b) 4500 L

(c) 1200 L (d) 7200 L
 17. A boy and girl together fill a cistern with water. The boy pours 4 litres of water every 3 minutes and the girl pours 3 litres of water every 4 minutes. How much time will it take to fill 100 litres of water in the cistern?

(a) 36 minutes (b) 42 minutes

(c) 48 minutes (d) 44 minutes
 18. Three pipes A, B and C can fill a cistern in 6 hours. After working at it together for 2 hours, C is closed and A and B fill it in 7 hours more. The time taken by C alone to fill the cistern is:
-

-
- (a) 14 hours (b) 15 hours
(c) 16 hours (d) 17 hours
19. A tap can fill an empty tank in 12 hours and another tap can empty half the tank in 10 hours. If both the taps are opened simultaneously, how long would it take for the empty tank to be filled to half its capacity?
(a) 10 hrs (b) 30 hrs
(c) 15 hrs (d) 20 hrs
20. Two pipes, P and Q can fill a cistern in 12 and 15 minutes respectively. Both are opened together, but at the end of 3 minutes, P is turned off. In how many more minutes will Q fill the cistern?
(a) 7 minutes (b) $7\frac{1}{2}$ minutes
(c) 8 minutes (d) $8\frac{1}{4}$ minutes
21. Pipes P and Q can fill a tank in 10 hours and 12 hours respectively and C can empty it in 6 hours. If all the three are opened at 7 am, at what time will one-fourth of the tank be filled?
(a) 10 am (b) 10 pm
(c) 11 pm (d) 11 am
22. If $\frac{3}{5}$ th of a cistern is filled in 1 minute, the time needed to fill the rest is:
(a) 40 sec (b) 30 sec
(c) 36 sec (d) 24 sec
23. A cylindrical cistern of diameter 25 cm is full of water. If 11 litres of water is drawn off, the water is drawn off, the water level in the cistern will drop by?
(use $\pi = \frac{22}{7}$)
(a) $10\frac{1}{2}$ cm (b) $12\frac{6}{7}$ cm
(c) $22\frac{2}{5}$ cm (d) $20\frac{2}{5}$ cm
24. If two pipes function simultaneously, a tank is filled in 12 hours. One pipe fills the tank 10 hours faster than the other. How many hours does the faster pipe alone take to fill the tank?
(a) 20 hrs (b) 18 hrs
(c) 15 hrs (d) 12 hrs
25. Two pipes X and Y can fill a cistern in 24 minutes and 32 minutes respectively. If both the pipes are opened together, then after how much time (in minutes) should Y be closed so that the tank is full in 18 minutes?
(a) 10 (b) 8
(c) 6 (d) 5
26. Two pipes A and B can fill a tank in 36 minutes and 45 minutes respectively. Another pipe C can empty the tank in 30 minutes. First A and B are opened. After 7 minutes, C is also opened. The tank is filled up in:
(a) 39 minutes (b) 46 minutes
(c) 40 minutes (d) 45 minutes
27. Having the same capacity 9 taps fill up a water tank in 20 minutes. How many taps of the same capacity are required to fill up the same water tank in 15 minutes?
(a) 10 (b) 12
(c) 15 (d) 18
28. A cistern is provided with two pipes A and B. A can fill it in 20 minutes and B can empty it in 30 minutes. If A and B be kept open alternatively for one minute each, how soon will the cistern be filled?
(a) 121 minutes (b) 110 minutes
(c) 115 minutes (d) 120 minutes
29. Pipe A can fill an empty tank in 6 hours and pipe B is 8 hours. If both the pipes are opened and after 2 hours pipe A is closed, how much time B will take to fill the remaining tank?
(a) $7\frac{1}{2}$ hours (b) $4\frac{3}{5}$ hours
(c) $2\frac{2}{5}$ hours (d) $3\frac{1}{3}$ hours
30. A water reservoir has two inlets and one outlet. Through the inlet it can be filled in 3 hours and 3 hours 45 minutes respectively. It can be emptied completely in 1 hour by the outlet. If the two inlets are opened at 01:00 pm and 02:00 pm respectively and the outlet at 03:00 pm then it will be emptied at?
(a) 05 : 55 pm (b) 05 : 00 pm
(c) 05 : 20 pm (d) 05 : 30 pm
-

Foundation

Solutions

1. (b); Time taken to fill the tank = 6 h.
Part of tank filled in 6 h = 1

$$\text{Part of tank filled in 1 h} = \frac{1}{6}$$

2. (d); Time taken to fill $\frac{1}{8}$ part of tank = 1 h
 $\therefore$ Time taken to fill the empty tank = 8 h.
3. (b); Time taken to empty a cistern = 3h

$$\begin{aligned} \text{Time taken to empty } \frac{2}{3} \text{ part of the cistern} \\ = 3 \times \frac{2}{3} = 2 \text{ h} \end{aligned}$$

4. (d); Given time taken by tap A to fill the tank = 60 min
Let Time taken by tap B to fill the tank = x min

$$\text{There, } \frac{1}{60} + \frac{1}{x} = \frac{1}{40} \Rightarrow \frac{1}{x} = \frac{1}{40} - \frac{1}{60} = \frac{3-2}{120}$$

$$\frac{1}{x} = \frac{1}{120} \Rightarrow x = 120 \text{ min}$$

5. (b); Given pipe A can fill a tank = 10 h
Pipe B can empty it = 6 h

$$\text{Time taken to empty the full tank} = \frac{1}{\frac{1}{6} - \frac{1}{10}}$$

$$= \frac{1}{\frac{5-3}{30}} = \frac{1}{\frac{2}{30}} = 15 \text{ h}$$

6. (b); Given,

Time taken by first tap to fill the tank (A) = 3 h
Time taken by second tap to fill the tank (B) = 4 h
And time taken to empty the full tank by third tap (C) = 5hr

$\therefore$ Part of the tank will be filled by all there taps in

$$1 \text{ hr} = \frac{1}{3} + \frac{1}{4} - \frac{1}{5} = \frac{20+15-12}{60} = \frac{23}{60} \text{ h}$$

$$\text{Required time} = \frac{60}{23} \text{ h} = 2\frac{14}{23} \text{ h}$$

7. (a); Given,

A = 30 min, B = 10 min, C = 40 min

$$\text{Part of the tank filled in 1 h} = \frac{1}{30} + \frac{1}{10} - \frac{1}{40}$$

$$= \frac{4+12-3}{120} = \frac{13}{120}$$

$$\text{Required time} = \frac{120}{13} = 9\frac{3}{13} \text{ h}$$

8. (b); Given, $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{10} \Rightarrow \frac{1}{a} = \frac{1}{30}, \frac{1}{b} = \frac{1}{40}$

$$\frac{1}{c} = \frac{1}{10} - \frac{1}{30} - \frac{1}{40} \Rightarrow \frac{12-4-3}{120} = \frac{5}{120}$$

$$c = 24 \text{ min}$$

9. (a); Given, a = 60, b = 75

$$\frac{1}{a} + \frac{1}{b} - \frac{1}{c} = \frac{1}{50} \Rightarrow \frac{1}{c} = \frac{1}{60} + \frac{1}{75} - \frac{1}{50}$$

$$= \frac{5+4-6}{300} \Rightarrow \frac{1}{c} = \frac{3}{300} \Rightarrow c = 100$$

10. (d); Let a, b, c be the time taken to fill the tank by pipes A, B and C respectively

$$\text{Given, } \frac{1}{a} + \frac{1}{b} = \frac{1}{6}, \frac{1}{b} + \frac{1}{c} = \frac{1}{10}, \frac{1}{a} + \frac{1}{c} = \frac{1}{12}$$

$$\text{Then, } 2\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = \frac{1}{6} + \frac{1}{10} + \frac{1}{12} = \frac{10+6+5}{60}$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{21}{120}$$

$$\text{Required time} = \frac{120}{21} = \frac{40}{7} = 5\frac{5}{7} \text{ h}$$

11. (b); Let 4x is the time taken to fill the tank by inlet B
Then x will be the time taken to fill tank by inlet A
 $\Rightarrow x = 15$

$$\text{Part of the tank will fill in 1 min} = \frac{4+1}{60} = \frac{1}{12} \text{ min}$$

$$\text{Required time} = 12 \text{ min}$$

12. (c); A = 16 h, B = 10 h

$$\text{Part of the tank filled in (2h)} = \frac{1}{16} + \frac{1}{10}$$

$$= \frac{5+8}{80} = \frac{13}{80}$$

$$\text{Part of the tank filled in (12 h)} = \frac{78}{80}$$

$$\text{Remaining part} = 1 - \frac{78}{80} = \frac{2}{80} = \frac{1}{40}$$

$$\frac{1}{40} \text{ part of the tank will be filled by inlet}$$

$$A = \frac{\frac{1}{40}}{\frac{1}{16}} = \frac{16}{40} = \frac{8}{20} = \frac{4}{10} = \frac{2}{5} \text{ hr}$$

$$\text{Required time} = 12 + \frac{2}{5} = 12\frac{2}{5} \text{ hr}$$

13. (d); 1 part of cistern will be empty by pipe A = 27 h

$$\frac{2}{3} \text{ part of cistern will be empty by pipe}$$

$$A = 27 \times \frac{2}{3} = 18 \text{ hr.}$$

14. (d); Given, A = 10 min, B = 6 min

$$\begin{aligned} \text{Part of tank will be empty in 1 min} &= \frac{1}{6} - \frac{1}{10} \\ &= \frac{5-3}{30} = \frac{1}{15} \end{aligned}$$

∴ Full tank will be empty in 15 min.

$$\therefore \frac{2}{3} \text{rd part of the tank will be empty in}$$

$$= 15 \times \frac{2}{3} = 10 \text{ minutes to empty the tank}$$

15. (c); Part of the tank will be filled in 1 h = $\frac{1}{8} - \frac{1}{16} = \frac{1}{16}$

Required time = 16 h

16. (b); Part of the tank that will be empty in 1 h = $\frac{1}{15} + \frac{1}{10}$

$$= \frac{2+3}{30} = \frac{5}{30} = \frac{1}{6} \Rightarrow \text{Required time} = 6 \text{ hrs}$$

17. (c); Given, A = 20 min, B = 25 min

In 1 min, part of the tank will be filled

$$= \frac{1}{20} + \frac{1}{25} = \frac{5+4}{100} = \frac{9}{100}$$

$$\text{In 5 min part of the tank will be filled} = \frac{45}{100}$$

$$\text{Remaining part} = 1 - \frac{45}{100} = \frac{55}{100} = \frac{11}{20}$$

It will take pipe A to fill the $\frac{11}{20}$ part of the tank

$$\begin{aligned} &\frac{11}{\frac{1}{20}} \\ &= \frac{20}{1} = 11 \text{ min} \end{aligned}$$

18. (c); By the formula

$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

$$\text{Here, } D_1 = 15, H_1 = 6, M_1 = 12$$

$$M_2 = ?, D_2 = 12, H_2 = 9$$

$$15 \times 6 \times 12 = x \times 12 \times 9$$

$$x = \frac{5 \times 6}{3} = 10$$

19. (c); Time taken by second pipe to fill the tank

$$= \frac{1}{\left(\frac{1}{8} - \frac{1}{12}\right)} = \frac{1}{\frac{3-2}{24}} = \frac{1}{\frac{1}{24}} = 24 \text{ hrs}$$

$$\text{Capacity of tank} = 24 \times 60 \times 6 = 8640 \text{ litres}$$

20. (a); Given,

First tap to fill the tank (A) = 8 hr

Second tap can empty the tank (B) = 16 hr

$$\text{Part of the tank will be filled in 1 h} = \frac{1}{8} - \frac{1}{16}$$

$$= \frac{2-1}{16} = \frac{1}{16} \Rightarrow \text{Required time} = 16 \text{ hr}$$

21. (c); Given,

First tap to fill the tank (A) = 5 hr

Second tap to empty the tank (B) = 4 hr

Part of the tank will be empty in 1 hr

$$= \frac{1}{4} - \frac{1}{5} = \frac{5-4}{20} = \frac{1}{20}$$

Required time = 20 hr

22. (a); Part of the tank will be filled in 1 hr

$$= \frac{1}{15} + \frac{1}{12} - \frac{1}{20} = \frac{4+5-3}{60} = \frac{6}{60} = \frac{1}{10}$$

Required time = 10 hr

23. (b); Part of tank will be emptied in 1 hr by 3rd pipe

$$= \frac{1}{60} + \frac{1}{70} - \frac{1}{60} = \frac{1}{70}$$

Required time = 70 min

24. (d); Let after x min the pipe be closed so that the tank will be filled full in half hour

$$\therefore \frac{30}{75} \times 2 + \frac{x}{45} = 1 \Rightarrow \frac{180+5x}{225} = 1$$

$$180 + 5x = 225 \Rightarrow 5x = 45 \Rightarrow x = 9 \text{ min}$$

25. (b); Let the required time = x

Part of the tank will be filled by tap (A and B)

$$= \frac{1}{60} + \frac{1}{40} = \frac{5}{120} = \frac{1}{24}$$

Then according to question

$$\frac{x}{2 \times 40} + \frac{x}{2 \times (24)} = 1 \Rightarrow \frac{x}{80} + \frac{x}{48} = 1 \Rightarrow x = 30$$

Required time = 30 min

26. (a); Part of tank filled in 1 min = $\frac{1}{15} + \frac{1}{18} = \frac{6+5}{90} = \frac{11}{90}$

Part of the tank that will be filled in 6 min

$$= \frac{66}{90} = \frac{11}{15}$$

Let x be the time taken to empty the tank by third tap.

$$\frac{\frac{11}{15}}{\frac{1}{x} - \frac{11}{90}} = \frac{33}{2} \Rightarrow \frac{90 - 11x}{90x} = \frac{11}{15} \times \frac{2}{33}$$

$$180x = 4050 - 495x \Rightarrow x = \frac{4050}{675} = 6 \text{ min}$$

27. (c); Part of the tank filled by all the three tap A, B, C in

$$1 \text{ hr} = \frac{1}{6}$$

$$\text{In 2 hr part of the tank will be fill} = \frac{1}{3}$$

$$\text{Remaining part} = 1 - \frac{1}{3} = \frac{2}{3}$$

Let a and b be time to fill the tank tap A and B respectively.

$$7 = \frac{\frac{2}{3}}{\frac{1}{a} + \frac{1}{b}} \Rightarrow \frac{1}{a} + \frac{1}{b} = \frac{2}{3} \times \frac{1}{7} = \frac{2}{21}$$

Part of the tank will be fill by the tap C 1 hr

$$= \frac{1}{6} - \frac{2}{21} = \frac{7-4}{42} = \frac{3}{42} = \frac{1}{14}$$

Required time = 14 hr.

28. (d); Part of the tank will be filled by all of the taps in 1

$$\text{min} = \frac{3+2-4}{12} = \frac{1}{12}$$

Required time = 12 hour

29. (b); Part of the tank will be empty in 1 hr = $\frac{1}{5} - \frac{1}{6} = \frac{1}{30}$

Required time = 30 hr

30. (c); By formula; $\frac{18 \times 5}{1440} = \frac{x \times 8}{1920}$

$$x = \frac{18 \times 5 \times 1920}{1440 \times 8} \Rightarrow x = 15 \text{ hour}$$

Moderate

1. (d); Part of the cistern that can be filled in 1 hr

$$= \frac{1}{14} + \frac{1}{16} = \frac{8+7}{112} = \frac{15}{112}$$

$$\text{Time taken to fill the cistern} = \frac{112}{15} \text{ hr}$$

$$= \frac{112}{15} \times 60 \text{ min} = 448 \text{ min}$$

Due to leakage, time taken to fill the cistern = 448 + 92 = 540 min = 9 hr

Time required to empty the tank

$$= \frac{1}{\frac{15}{112} - \frac{1}{9}} = \frac{112 \times 9}{23} = 43 \frac{19}{23} \text{ h}$$

2. (c); Part of the tank that will be filled when pipe

$$\text{P and R are opened for 1 min} = \frac{1}{12} + \frac{1}{15} = \frac{9}{60}$$

∴ Part of the tank that will be filled after 5 min

$$= \frac{45}{60} = \frac{3}{4}$$

6 min is the time to empty the tank by tap M.

$$\therefore \text{Required time} = \frac{3}{4} \times \frac{1}{\left(\frac{1}{6} - \frac{9}{60}\right)}$$

$$= \frac{3}{4} \times \frac{60}{1} = 45 \text{ min}$$

3. (c); Required time = $\frac{40}{4} \text{ min} = 10 \text{ min}$

4. (a); Let first pipe fill the tank in (x) h then
Second pipe will fill it in (x + 10)h

$$\frac{1}{x} + \frac{1}{x+10} = \frac{1}{12} \Rightarrow \frac{x+10+x}{x(x+10)} = \frac{1}{12}$$

$$\frac{2x+10}{x^2+10x} = \frac{1}{12} \Rightarrow 24x + 120 = x^2 + 10x$$

$$x^2 - 14x - 120 = 0$$

$$x^2 - 20x + 6x - 120 = 0$$

$$x(x-20) + 6(x-20) = 0 \Rightarrow x = 20$$

5. (b); Let total time = x

$$\therefore \frac{2}{15} + \frac{x}{20} = 1 \Rightarrow \frac{x}{20} = 1 - \frac{2}{15}$$

$$\frac{x}{20} = \frac{13}{15} \Rightarrow x = \frac{52}{3} \text{ min}$$

6. (d); Part of the tank that will be filled by both pipe A

$$\text{and B in 1 min} = \frac{1}{24} + \frac{1}{32} = \frac{4+3}{96} = \frac{7}{96}$$

Let after x min, pipe B should be closed

$$\therefore x \times \frac{7}{96} + \frac{(9-x)}{24} = 1$$

$$\frac{7x+36-4x}{96} = 1$$

$$3x + 36 = 96 \Rightarrow 3x = 60 \Rightarrow x = 20 \text{ min}$$

7. (c); Part of the tank will be filled when all of the pipes A, B and C are opened alternatively, each of 1 min (3 min)

$$= \frac{1}{20} + \frac{1}{30} - \frac{1}{15} = \frac{3+2-4}{60} = \frac{1}{60}$$

$$\therefore \text{Time taken to fill} \left[1 - \left(\frac{1}{20} + \frac{1}{30} \right) \right] \text{ Or}$$

$$\frac{55}{60} \text{ th part of the tank filled in } = 3 \times 55 = 165 \text{ min}$$

$$\therefore \text{Remaining part} = 1 - \frac{55}{60} = \frac{1}{12}$$

$$\text{Tank A fills } \frac{1}{20} \text{ part in 1 min}$$

$$\therefore \text{Remaining part} = \frac{1}{12} - \frac{1}{20} = \frac{1}{30}$$

$$\text{i.e., } \frac{1}{30} \text{ th part is filled by B in 1 min}$$

$$\therefore \text{Total time} = (165 + 1 + 1) = 167 \text{ min}$$

8. (d); When taps are not opened properly

$$\text{Tap A will fill the tank} = 25 \times \frac{6}{5} = 30 \text{ min}$$

$$\text{Tap B will fill the tank} = 20 \times \frac{3}{2} = 30 \text{ min}$$

$$\therefore \text{Part of the tank will be filled in 1 min} = \frac{1}{30} + \frac{1}{30}$$

$$= \frac{1}{15}$$

$$\therefore \text{Required time} = 15 \text{ min}$$

9. (b); Part of the tank will be filled in 1 min

$$= \frac{1}{20} + \frac{1}{15} + \frac{1}{12} = \frac{3+4+5}{60} = \frac{12}{60} = \frac{1}{5}$$

$$\therefore \text{Required time} = 5 \times \frac{40}{100} = 2 \text{ min}$$

10. (d); Part of the tank will be filled in 1 min by tap

$$\text{A and B} = \frac{1}{4} + \frac{1}{5} = \frac{5+4}{20} = \frac{9}{20}$$

In 2 min. part of the tank will be filled

$$= 2 \times \frac{9}{20} = \frac{9}{10}$$

Time Required by third pipe to empty the tank

$$= \frac{9}{10} \times \frac{1}{\left(\frac{1}{2} - \frac{1}{4} - \frac{1}{5} \right)} = \frac{9}{10} \times 20 = 18 \text{ min}$$

11. (c); In 1 hr the amount of water that will be filled in the tank = 42 + 56 - 48 = 50 L

$$\therefore \text{Capacity of the tank} = 50 \times 16 = 800 \text{ L}$$

12. (c); In 1 min the boy will pour water in tank = $\frac{4}{3}$ L

$$\text{In 1 min the girl will pour water in tank} = \frac{3}{4} \text{ L}$$

$$\therefore \text{Both will pour water in 1 min} = \frac{1}{\frac{4}{3} + \frac{3}{4}} = \frac{12}{25}$$

$$\therefore \text{Required time} = \frac{12}{25} \times 100 = 48 \text{ min}$$

13. (a); Let inlet pipe fill the tank in x hr

$\therefore$ Outlet pipe will empty the tank in 2x hr

$$\therefore \frac{1}{x} - \frac{1}{2x} = \frac{1}{x+36} \Rightarrow \frac{1}{2x} = \frac{1}{x+36} \Rightarrow x = 36 \text{ hr}$$

14. (a); Diameter $\Rightarrow 1 \quad 3 \quad 4$

$$\text{Efficiency} \Rightarrow 1 : 9 : 16 \rightarrow 26 \rightarrow 16 \text{ min}$$

$$\downarrow$$

$$\frac{16 \times 26}{16}$$

Thus, third pipe alone will fill the tank in 26 min.

15. (c); Part of the tank will be filled in 1 hr

$$= \frac{1}{36} + \frac{1}{45} = \frac{9}{180} = \frac{1}{20}$$

∴ Required time = 20 hr

16. (c); In 1 minute the pipe of 2 cm fills

$$\frac{1}{61} \times \frac{1}{4} \text{ of the cistern}$$

In 1 min the pipe of $1\frac{1}{3}$ cm diameter fill $\frac{1}{61} \times \frac{4}{9}$ of the cistern

In 1 min the pipe of 1 cm diameter fill $\frac{1}{61}$ of the cistern.

$$\begin{aligned} \therefore \frac{1}{61 \times 4} + \frac{1}{61} \times \frac{4}{9} + \frac{1}{61} &= \frac{9+16+36}{9 \times 4 \times 61} \\ &= \frac{61}{36 \times 61} = \frac{1}{36} \end{aligned}$$

∴ The whole is filled with 36 minutes.

17. (c); Part of the tank will be filled, by both pipe in 1 hr

$$= \frac{1}{20} + \frac{1}{30} = \frac{5}{60} = \frac{1}{12}$$

∴ In 1 hr the leak will empty the part of tank

$$= \frac{1}{12} \times \frac{1}{3} = \frac{1}{36}$$

And to fill $\frac{1}{3}$ of the tank, both pipe will take

$$= \frac{12}{3} = 4 \text{ hr.}$$

to fill other $\frac{2}{3}$ part of the tank, it will take

$$= \frac{2}{3} \times \frac{1}{\frac{1}{20} + \frac{1}{30} - \frac{1}{36}} = \frac{2}{3} \times \frac{1}{\frac{9+6-5}{180}}$$

$$= \frac{2}{3} \times \frac{180}{10} = 12 \text{ hr}$$

∴ Total time = (12 + 4) hr = 16 hr

18. (b); Part of tank will be full in 1 hr

$$= \frac{1}{4} + \frac{1}{6} - \frac{1}{8} = \frac{6+4-3}{24} = \frac{7}{24}$$

$$\therefore \text{Required time} = \frac{24}{7} = 3\frac{3}{7} \text{ hrs.}$$

19. (b); Let first pipe can fill the reservoir in x hr

∴ Second pipe will fill it in (x + 5) hr

$$\therefore \frac{1}{x} + \frac{1}{x+5} = \frac{1}{6} \Rightarrow \frac{x+5+x}{x(x+5)} = \frac{1}{6}$$

$$6(2x+5) = x^2+5x \Rightarrow 12x+30 = x^2+5x$$

$$x^2-7x-30=0 \Rightarrow x^2-10x+3x-30=0$$

$$(x+3)(x-10)=0 \Rightarrow x=10 \text{ hrs}$$

20. (b); Let x, y and z be time to fill the cistern by pipes, A, B and C respectively. The part of the tank will be filled in 2 hr by all the pipe

$$= 2 \times \left(\frac{1}{6}\right) = \frac{1}{3}$$

$$\text{Remaining part} = 1 - \frac{1}{3} = \frac{2}{3}$$

∴ Time taken by pipe A and B together to fill the

$$\text{tank} = 8 \times \frac{3}{2} = 12 \text{ h}$$

∴ by pipe C alone to fill the tank = $\frac{1}{6} - \frac{1}{12} = \frac{1}{12}$

∴ Required time = 12 hr

Difficult

1. (d); Let x be the time, after that B be closed so that the tank is full in 18 min.

$$\therefore \frac{18}{24} + \frac{x}{32} = 1 \Rightarrow \frac{72+3x}{96} = 1$$

$$72+3x=96 \Rightarrow 3x=24 \Rightarrow x=8 \text{ min}$$

2. (c); Part of the tank will be empty in 1 hr

$$= \frac{1}{10} + \frac{1}{15} - \frac{1}{12} = \frac{6+4-5}{60} = \frac{5}{60} = \frac{1}{12}$$

∴ Required time = 12 hr

3. (d); In 30 min the part of the tank will be filled by

$$\text{both tap} = \frac{30}{36} = \frac{5}{6}$$

$$\therefore \text{Required part} = 1 - \frac{5}{6} = \frac{1}{6}$$

∴ $\frac{1}{6}$ part of the tank will be filled by pipe A in 10 min

∴ tap A will take 60 min. to fill the tank

∴ Pipe B will take time to fill the tank

$$= \frac{1}{36} - \frac{1}{60} = \frac{5-3}{180} = \frac{2}{180} = \frac{1}{90} ;$$

Hence B will take 90 min to fill the tank.

4. (d); The part of tank that will filled in

$$2 \text{ min} = \frac{1}{45} - \frac{1}{60} = \frac{4-3}{2} = \frac{1}{180}$$

Part of the tank $\left(1 - \frac{1}{45} = \frac{44}{45}\right)$ will be filled in

$$= \frac{176}{180} \times 180 \times 2 = 352 \text{ min}$$

And $\frac{1}{45}$ of the tank will be filled in 1 min

$$\therefore \text{Total time} = (352 + 1) \text{ min} \\ = 353 \text{ min} = 5 \text{ hr } 53 \text{ min}$$

5. (d); Let time = x

$$\therefore \frac{5}{20} + \frac{(x+5)}{60} = 1 \Rightarrow \frac{15+x+5}{60} = 1$$

$$20 + x = 60 \Rightarrow x = 40 \text{ min}$$

6. (b); (A's 1 hour work) + (A + B)'s 1 hour work

$$\frac{1}{3} + \left(\frac{1}{3} + \frac{1}{4}\right) = \frac{11}{12}$$

At 5'o' clock $\frac{11}{12}$ part of the cistern is filled.

Total discharge if all the pipes opened

$$= 1 - \left(\frac{1}{3} + \frac{1}{4}\right) \Rightarrow \frac{12 - (3+4)}{12} = \frac{5}{12}$$

Time required to empty $\frac{11}{12}$ part of the tank

$$= \frac{\frac{11}{12}}{\frac{5}{12}} = \frac{11}{5} = 2.2 \text{ hr.}$$

Hence required time = 7.12 PM

7. (c); Part of the tank will be filled in 1 min

$$= \frac{1}{30} + \frac{1}{20} + \frac{1}{10} = \frac{2+3+6}{60} = \frac{11}{60}$$

In 3 min the part of the tank will be filled = $\frac{33}{60} = \frac{11}{20}$

Part of the solution R in the liquid after

$$3 \text{ min} = \frac{\frac{3}{10}}{\frac{11}{20}} = \frac{60}{110} = \frac{6}{11}$$

8. (b); Part of tank will be filled by both pipe A and B in

$$1 \text{ hr} = \frac{2}{3} \times \frac{1}{8} = \frac{1}{12}$$

$$\therefore \text{Required time} = \frac{12}{2-1} = 12 \text{ hr}$$

9. (d); Part of the tank that will fill the tank in 1 hr by all

$$\text{these pipes} = \frac{1}{2} \left(\frac{1}{6} + \frac{1}{10} + \frac{2}{15} \right) = \frac{1}{5}$$

So, tank will be filled in 5 h

10. (a); Let tank is filled in x hours

$$\frac{x}{15} + \frac{x}{20} - \frac{10}{25} = 1 \Rightarrow \frac{20x+15x-120}{300} = 1$$

$$35x = 300 + 120 \Rightarrow x = \frac{420}{35} = 12$$

Therefore tank is filled in 12 hours

11. (d); (A's 1 hour work) + (A + B)'s 1 hour work

$$= \frac{1}{10} + \left(\frac{1}{10} + \frac{1}{12}\right) = \frac{17}{60}$$

$$\text{Remaining part} = 1 - \frac{17}{60} = \frac{43}{60}$$

Now (A + B + C)'s 1 hour work

$$= \frac{1}{10} + \frac{1}{12} + \frac{1}{15} = \frac{1}{4}$$

$$\frac{1}{4} \text{ part is filled by then in } \left(4 \times \frac{43}{60}\right) = 2 \text{ hr } 52 \text{ min}$$

$$\therefore \text{Total time} = 2 + 2 \text{ hr } 52 \text{ min} = 4 \text{ hr } 52 \text{ min}$$

12. (a) Part of the tank that will filled when all the pipes

$$\text{are open in 3 min} = \frac{1}{20} + \frac{1}{30} - \frac{1}{15} = \frac{3+2-4}{60} = \frac{1}{60}$$

$$\left[\left(1 - \frac{1}{20} - \frac{1}{30} \right) = \frac{55}{60} \right] \text{ part of the tank will be filled in } = 55 \times 3 = 165 \text{ min}$$

$$\text{Remaining part} = 1 - \frac{55}{60} = \frac{1}{12}$$

$$\text{Tap A fills } \frac{1}{20} \text{ part in 1 min}$$

$$\text{Remaining part} = \frac{1}{12} - \frac{1}{30} = \frac{1}{30}$$

$$\frac{1}{30} \text{ th part is filled by B in 1 min.}$$

$$\therefore \text{Total time} = (165 + 1 + 1) = 167 \text{ min.}$$

13. (c); Let A can fill the tank = x

$$\therefore \text{B will fill the tank} = \frac{x}{2}$$

$$\therefore \text{C will fill the tank} = \frac{x}{4}$$

$$\therefore \frac{4}{x} + \frac{2}{x} + \frac{1}{x} = \frac{1}{5} \Rightarrow \frac{7}{x} = \frac{1}{5} \Rightarrow x = 35 \text{ hrs}$$

14. (d); Let total time = x

$$\therefore \frac{4}{15} + \frac{x}{20} = 1 \Rightarrow \frac{16+3x}{60} = 1 \Rightarrow 3x = 60 - 16$$

$$x = \frac{44}{3} \text{ min} \Rightarrow = \frac{44}{3} \times 60 \text{ sec} = 880 \\ = 14 \text{ min } 40 \text{ sec}$$

15. (d); In one hour the part of the tank will be filled

$$= \frac{1}{12} + \frac{1}{15} = \frac{5+4}{60} = \frac{9}{60}$$

In second hour part of the tank will be filled

$$= \frac{1}{12} + \frac{1}{20} = \frac{5+3}{60} = \frac{8}{60}$$

$\therefore$ In two hour part of the tank will be filled

$$= \frac{9}{60} + \frac{8}{60} = \frac{17}{60}$$

$$\text{In 6 hour part of the tank will be filled} = \frac{51}{60}$$

$$\text{Remaining part} = 1 - \frac{51}{60} = \frac{9}{60} = \frac{3}{20}$$

$$\therefore \text{Remaining part will be filled in} = \frac{\frac{3}{20}}{\frac{3}{60}}$$

$$= \frac{3}{20} \times \frac{60}{3} = 1 \text{ hr}; \therefore \text{Total time} = (6 + 1) = 7 \text{ hr.}$$

16. (a); Let the filling capacity of the tank = x m³/min
Emptying capacity of the tank = (x + 10) m³/min

$$\therefore \frac{2400}{x} - \frac{2400}{(x+10)} = 8 \Rightarrow x^2 + 10x - 3000 = 0$$

$$(x-50)(x+60) = 0 \Rightarrow x = 50 \text{ m}^3/\text{min}$$

17. (d); Work done by the inlet in 1 hr

$$= \frac{1}{8} - \frac{1}{12} = \frac{1}{24}$$

Work done by the inlet in 1 min

$$= \frac{1}{24} \times \frac{1}{60} = \frac{1}{1440}$$

$$\text{Volume of } \frac{1}{1440} \text{ part} = 6 \text{ litres}$$

$$\text{Volume of tank} = 6 \times 1440 = 8640 \text{ litres}$$

18. (c); Part of the tank will be empty by waste pipe in

$$1 \text{ min} = \frac{1}{20} + \frac{1}{24} - \frac{1}{15} = \frac{6+5-8}{120} = \frac{3}{120} = \frac{1}{40}$$

$$\text{capacity of tank} = 40 \times 3 = 120 \text{ gallons.}$$

19. (c); Part of the tank will be filled in = 2 hr = $\frac{2}{6} = \frac{1}{3}$

$$\therefore \text{Remaining part} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\therefore \text{together (A + B) can fill the tank} = \frac{7}{2} \times 3 = \frac{21}{2}$$

$$\therefore \text{C's 1 hour work} = \frac{1}{6} - \frac{2}{21} = \frac{1}{14}$$

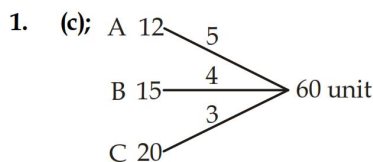
$$\text{Required time} = 14 \text{ hrs}$$

20. (a); Let after x minute the pipe B should be closed

$$\frac{20}{24} + \frac{x}{48} = 1 \Rightarrow \frac{40+x}{48} = 1$$

$$40 + x = 48 \Rightarrow x = 8$$

Previous Year (Memory Based)



In 1 hour, part of tank that can be filled
= 5 + 4 = 9 units

In 2 hour, part of tank that can be filled
= 5 + 3 = 8 units

In 6 hour, part of tank that can be filled
= 17 × 3 = 51 units

left part = (60 - 51) = 9 units

Which will be filled in 1 hour by pipe A and B.

$\therefore$ Total time = 7 hours

2. (b); 'd' diameter : '2d' diameter
Efficiency 1 : 4

$$\therefore \text{Required time} = \frac{40}{4} = 10 \text{ min.}$$

3. (b); Required time = $\frac{1}{\frac{1}{3} - \frac{2}{7}} = \frac{1}{\frac{7-6}{21}} = 21$ hours

4. (b); Required time = $\frac{1}{\frac{1}{60} + \frac{1}{75} - \frac{1}{50}}$
 $= \frac{300}{5+4-6} = 100$ minutes

5. (d); Remaining part after $\frac{1}{2}$ hour = $1 - \frac{1}{2} = \frac{1}{2}$

to fill $\frac{1}{2}$ part, time taken = $\frac{\frac{1}{2}}{\frac{3}{4}} = \frac{3}{4}$ hr

$\therefore$ Required time = $3\frac{3}{4}$ hr

3 hr 45 min.

6. (c);

	Faster	:	Slower
Efficiency	3	:	1
Time	1	:	3

$= \frac{1}{x} + \frac{1}{3x} = \frac{1}{36}$

$\frac{3+1}{3x} = \frac{1}{36}$

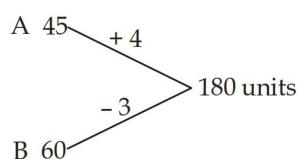
$x = 48$ minutes

$\therefore$ Required time = $48 \times 3 = 144$ minutes

7. (b); both pipes will empty the tank in = $\frac{1}{\frac{1}{5} - \frac{1}{8}}$

$= \frac{1}{\frac{8-5}{40}} = \frac{40}{3}$ hrs

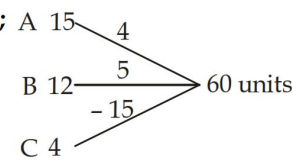
$\therefore$ Required time = $\frac{\frac{3}{4}}{\frac{3}{40}} = 10$ hours

8. (d); 

In 2 minutes part of tank will be filled = 4.3 = 1 units

To fill $(180 - 4 = 176)$ units of tank, time = $176 \times 2 = 352$ units.

$\therefore$ Required time = $352 + 1 = 353$ minutes = 5 hr 53 minutes

9. (d); 

At 11 A.M. part of tank filled = $4 \times 3 + 5 \times 2 = 22$ units

In 1 hr, all the three taps will empty the tank = $15 - 5 - 4 = 6$ units

To empty 6 units, time = $\frac{22}{6} = 3$ hr 40 units

$\therefore$ Required time = 02 : 40 PM

10. (d); Required time = $\frac{1}{\frac{1}{x} - \frac{1}{y}} = \frac{xy}{y-x}$ hours

11. (c); $12 \times 6 \times 15 = x \times 9 \times 12$
 $x = 10$

12. (a);

Efficiency	2	:	1
Time	1	:	2

$2x = x - 36 \Rightarrow x = 36$ hours

13. (d); $\frac{1}{A} + \frac{1}{B} = \frac{1}{3C}$... (i)

$\frac{40}{A} + \frac{30}{B} = 1$... (ii)

equation (i) $\times 40$ - equation (ii)

$\frac{40}{A} + \frac{40}{B} = \frac{40}{36}$

$\frac{40}{A} + \frac{30}{B} = 1$

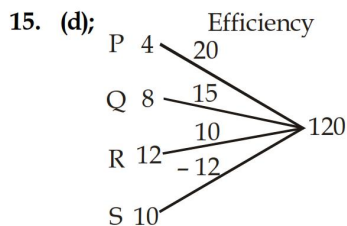
$\frac{10}{B} = \frac{40}{36} - 1$

$\frac{10}{B} = \frac{4}{36} \Rightarrow B = 90$ minutes

14. (c); Time taken to empty the tank by pipe

$c = \frac{1}{\frac{1}{20} + \frac{1}{24} - \frac{1}{15}} = \frac{120}{6+5-8} = 40$ minutes

$\therefore$ Capacity of tank = $40 \times 3 = 120$ gallons



- (A) Q efficiency 15 units/hr
 (B) (P + R - S) efficiency = 30 + 10 - 12 = 28 unit / hr
 (C) (P + S) efficiency = 30 - 12 = 18 unit/hr
 (D) (P + Q - S) efficiency = 30 + 15 - 12 = 33 unit/hr
 $\therefore$ efficiency of D is highest

16. (d); $\frac{1}{10} - \frac{1}{x} = \frac{1}{15}$

$$\frac{1}{x} = \frac{1}{10} - \frac{1}{15} = \frac{3-2}{30}$$

$$x = 30 \text{ hours}$$

$$\therefore \text{Required capacity of tank} = 30 \times 4 \times 60 = 7200 \text{ L}$$

17. (c); Boy = $\frac{3}{4}$, Girl = $\frac{4}{3}$

$$\text{Boy} + \text{Girl} = \frac{16+9}{12} = \frac{25}{12} \text{ litres}$$

$$\therefore \frac{25}{12} \text{ litre} \rightarrow 1 \text{ minute}$$

$$100 \text{ litre} \rightarrow \frac{1}{25} \times 12 \times 100 = 48 \text{ minutes}$$

18. (a); After 2 hours, part filled = $\frac{2}{6} = \frac{1}{3}$

$$\text{Remaining part} = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\therefore \frac{\frac{2}{3}}{\frac{1}{A} + \frac{1}{B}} = 7$$

$$\frac{1}{A} + \frac{1}{B} = \frac{2}{21}$$

$$\therefore \text{Required time} = \frac{1}{\frac{1}{6} - \frac{2}{21}} = \frac{42}{7-4} = 14 \text{ hours}$$

19. (c); Required time = $\frac{1}{2} \left[\frac{1}{\frac{1}{12} - \frac{1}{20}} \right]$

$$= \frac{1}{2} \times \frac{60}{5-3} = 15 \text{ hrs.}$$

20. (d); $\frac{3}{12} + \frac{3+x}{15} = 1 \Rightarrow \frac{15+12+4x}{60} = 1$

$$4x = 60 - 27 \Rightarrow 4x = 33$$

$$x = \frac{33}{4} \Rightarrow x = 8\frac{1}{4} \text{ minutes}$$

21. (b); Time = $\frac{1}{4} \left[\frac{1}{\frac{1}{10} + \frac{1}{12} - \frac{1}{6}} \right]$

$$= \frac{1}{4} \times \frac{60}{6+5-10} = 15 \text{ hrs.}$$

$$\therefore \text{Required time} = 10 \text{ PM}$$

22. (a); $\frac{3}{5} \text{ th} \rightarrow 1$

$$1 \rightarrow \frac{5}{3} \Rightarrow \frac{2}{5} \rightarrow \frac{5}{3} \times \frac{2}{5} = \frac{2}{3} \text{ minutes}$$

$$= 40 \text{ sec.}$$

23. (c); Let it will drop by h cm

$$\therefore \frac{22}{7} \times \frac{25}{2} \times \frac{25}{2} \times h = 11000$$

$$h = 22\frac{2}{5} \text{ cm}$$

24. (a); $\frac{1}{x} + \frac{1}{x+10} = \frac{1}{12}$

$$\frac{x+10+x}{x(x+10)} = \frac{1}{12}$$

$$(2x+10)12 = x^2 + 10x$$

$$24x + 120 = x^2 + 10x$$

$$x^2 - 14x - 120 = 0$$

$$x^2 - 20x + 6x - 120 = 0$$

$$x(x-20) + 6(x-20) = 0$$

$$(x-20)(x+6) = 0$$

$$x = 20 \text{ hrs.}$$

25. (b); $\frac{18}{24} + \frac{x}{32} = 1 \Rightarrow \frac{72+3x}{96} = 1$

$$72 + 3x = 96$$

$$3x = 24 \Rightarrow x = 8 \text{ minutes}$$

26. (b); $\frac{x}{36} + \frac{x}{45} - \frac{(x-7)}{30} = 1$

$$\frac{5x + 4x - 6x + 42}{180} = 1$$

$$3x = 138$$

$$x = 46 \text{ minutes}$$

27. (b); $9 \times 20 = x \times 15$

$$x = \frac{9 \times 4}{3} = 12$$

28. (c);

$$\begin{array}{rcl} \text{A } 20 & +3 & \\ & & \searrow \\ & & 60 \text{ units} \\ & \nearrow & \\ \text{B } 30 & -2 & \end{array}$$

In 2 minutes, tank filled = $3 - 2 = 1$ units

In 114 minutes tank filled = 57 units

Remaining = $60 - 57 = 3$ units

Which will be filled in 1 minute by tap A

$\therefore$ Total time = $114 + 1 = 115$ minutes

29. (d); After 2 hours, Remaining part = $1 - \frac{14}{24}$

$$= \frac{10}{24} = \frac{5}{12}$$

$$\therefore \text{ Required time} = \frac{\frac{5}{12}}{\frac{1}{8}} = \frac{40}{12}$$

$$= \frac{20}{6} = \frac{10}{3} = 3\frac{1}{3} \text{ hours}$$

30. (c);

$$\begin{array}{rcl} \text{Inlet } 180 & +5 & \\ \text{Inlet } 225 & +4 & \\ & & \searrow \\ & & 900 \text{ unit} \\ & \nearrow & \\ \text{Outlet } 60 & -15 & \end{array}$$

At 3 PM, tank filled = $5 \times 2 + 4 = 14$ units

14 units of tank can be embtied in = $\frac{14}{6}$

= 2 hrs 20 min

$\therefore$ Required time = 5 : 20 PM